

Designing Project Management Systems

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Abstract

The purpose of this paper is to develop the concept of project management system from the perspective of systems science. There is a need to extend the body of knowledge for project management. In particular, the application of systems perspectives and systems theory offers a significant opportunity to advance the current state of project management knowledge. Although there have been suggestions of systems approaches for project management, rigorous systems science has not been used to support these depictions. First, we develop the background and define a perspective of project management systems. Second, assumptions and principles are drawn from systems science to provide a foundation for project management systems. Third, a model for project management systems is developed from systems sciences and management cybernetics. Our initial explorations are promising, demonstrated by a case study application review using a complex system of systems (SoS) project encompassed multiple government agencies, and we offer future directions and implications for further model refinements, applications, and research into project management systems.

1. Introduction

Project Management is a recognized discipline focused on facilitating the structured accomplishment of work. The fundamental concept of project management is given by the Project Management Institute (PMI) as, "The application of knowledge, skills, tools, and techniques to project activities to meet project requirements" [1, p.383].

The relative importance of the project management discipline can be seen in the exponential rise in the membership of the Project Management Societies. For instance, International Project Management Association (IPMA), as being the oldest association on project management, has started in 1965, and today's membership is currently exceeding 40,000 members in 40 countries [2].

Project management continues to evolve as a discipline and is here to stay. The expansion of project management is certainly a phenomenon of interest. The mid-1990's Lewis [3] estimated that over half of the work in organizations was being accomplished within some framework of a project. At a most basic level, project management is recognized in the literature as an approach for effectively providing a unique product or service within a specified time [3], [4] and [5]. Cleland [4, p.8] defines a project as "...any undertaking that has a defined objective, a cost parameter, and a time element for its deployment." However, the project management discipline has also recognized the need to advance the body of knowledge as well as the role of research in facilitating that advancement [6].

There is a need to extend the body of knowledge for project management. In particular, the application of systems perspectives and systems theory offers a significant opportunity to advance the current state of project management knowledge. The application of a systems perspective to project management has generated several representations of project management systems and calls for systems based approaches to project management [7], and [8]. In fact, the term "integration" has been used to capture the need to focus on processes to provide for coordination of all the elements of a project [1] and [7]. In a recent study, Koltveit et al. [42] examines project management perspectives under six categories: the task, the leadership, the system, the stakeholder, the transaction cost, and the business by project perspective. The investigation of these various perspectives in articles between 1983 and 2004 of the *International Journal of Project Management* yields that only 4.7% of observations are related to system perspective [42]. The rigorous application of systems science to project management has been lacking. *The primary purpose of this paper is to develop a systems science based model for project management systems.*

The paper is organized to present: (1) an explanation of the current work and establish an initial definition of "project management system", (2) essential assumptions for project management systems from a

systems science perspective, (3) essential systems principles supporting development of a model for project management systems, (4) an introductory overview of the model for project management systems, and (5) a case study application review. The paper closes with implications for further refinement of the model, application development challenges, and future research directions for development of project management systems.

2. Project Management Systems

The systems science view of designing, analyzing, and operating complex systems has successfully developed over several decades [9], [10] and [11]. Although system science has a long history, it has seen limited application to project management. However, there have been references to the application of “systems” concepts and project management systems. For instance, Kerzner [12] discusses the notion of “systems” as applied to project management as well as suggesting a “systems approach”. In addition, Lewis [13] suggests the importance of applying “systems thinking” for more effective project management while Laufer et al. [14] suggests systems principles such as simple procedures for projects.

The term “*project management system*”, hereafter referred to as PMS, has been applied to different industries to guide [15] and assess performance [16] of projects. Metcalfe [17] addresses the social and organizational aspects underlying PMSs, which are developed as systems to provide management information. Yolles and Pirani [18] used the term PMS as it applied expert systems to assist project management training and support. Laufer et al. [14] suggest intensive communications, simple procedures, and systematic monitoring as the system focus for project management. Finally, Cleland [7] describes several subsystems necessary for a PMS. Although systems have been recognized as important to project management, rigorous systems science development of PMSs has not received significant attention.

Based on the descriptions of PMSs in the literature, we observe the following: (1) lack of a substantive definition or agreement of what constitutes a PMS, (2) insufficient depth or linkage to the underlying systems (theory), principles, or thinking to support the formulation of the PMS, (3) limited elaboration of PMS frameworks in sufficient detail to permit closer scrutiny, critique, and examination, (4) limited guidance, or methodology, to design, assess, or transform PMSs, and (5) absence of empirical research to better understand the phenomena associated with PMSs.

To amplify the concept of PMS, we define a PMS as:

The structured set of technical and human entities that interact both formally and informally within a specific context to produce project results. The products of interaction are patterns of decision, action, and interpretation that drive project performance.

3. Systems science foundations

The application of systems science to project management requires some background to develop an appropriate foundation. In this section we introduce four fundamental assumptions for application of systems science to development of a model for PMSs.

A first assumption for the systemic view of PMSs suggests that *every project has a unique PMS*. This system may be tacit or explicit, but it is responsible for the results of the project. Ask most project managers to identify their PMS and the responses are likely to range from a two-page paper (actually provided by one company as their PMS) to an acknowledgement that they have no representation of their PMS. This is not to suggest that projects cannot be successful without an explicit understanding of the supporting PMS. On the contrary, explicit representation provides a basis for analyses, evaluation, and systemic transformation that increases the probability for success. Understanding the technical, contextual, and human dimensions of a PMS would minimally identify systemic deficiencies and permit targeting of scarce resources to high value adding areas.

Our second systems assumption holds that *there is not one correct representation of a PMS*. Instead, there are multiple perspectives, project system configurations, and representations that may emerge to depict a PMS. Therefore, although each project will have a unique PMS, so will there be varying interpretations of what constitutes that particular PMS. The need for inclusion of multiple perspectives [9] has long been recognized as an important consideration in systems-based methodologies. In addition, interpretations of situations and systems are influenced by language, worldview, and perceptions [20], [21], [22] and [23]. This supports the notion that a PMS is dynamic and subject to reformulation over time and across individual perspectives.

Based on new knowledge, shifts in perceptions, and changing conditions, the potential exist for multiple PMS representations for the same project. Consequently, it is inappropriate to think that a PMS will be static and not capable of shifting over the life of the project. There is no absolutely ‘correct’ or ‘constant’ depiction of a PMS. This echoes the importance of dialogue, interpretation, and shared understanding that is prevalent in the systems literature, [24], [25] and [26]. In effect, a PMS is arbitrary, depending on those providing the

representation. This does not dismiss the responsibility for PMS development but only serves to accentuate the necessity for continuous dialogue in evolving the system representation.

The third assumption suggests that *project problems are a product of the underlying 'PMS'*. This underlying system produces the often 'symptomatic' conditions labeled as project problems. Therefore, it is inconsequential to talk of project problems, or the manifest conditions defined as problematic, as separate and distinct from the contributing PMS responsible for producing the condition(s). This assumption is consistent with system-based problem solving approaches that recognize the complex systems nature of problems [20], [27], [28] and [29]. Attention must be focused on the aspects of the PMS that are producing the undesirable conditions. In essence, project problems must first be examined to determine if they represent a faulty PMS in disguise.

Finally, the fourth assumption asserts that deployed PMSs will produce intended as well as unintended consequences. Since system design is not precise, the consequences of deploying, or modifying an existing PMS (structure, relationships, patterns), will produce effects that cannot be predicted in advance. Therefore, the design for effective PMSs must include the provision for continuing modification of the system based on initial or latent unintended consequences.

Given the four systems view assumptions we can move forward to examining essential systems principles that provide a further foundation for PMSs.

3.1 Systems principles for PMSs. In taking a systems perspective, to sharpen our focus, we submit five fundamental systems concepts that influenced development of the PMS model.

3.1.1 System purpose – Beer, [30] and [31], has proposed that the purpose of a system is 'what it does'. In considering the implications for PMSs, we restate the point as 'a system can only produce what it is capable of producing.' Capability is a function of both design and execution of the system. The resulting system outputs (patterns, products, and services) and outcomes (impacts) of the system in operation are 'the' products of the system, regardless as to whether or not we have a complete understanding of the system.

The systems perspective of design does not confuse intention with results. Regardless of the well-meaning intentions for a system design, purpose is based on 'actual' results, not those that were 'intended'. Therefore, it is a fundamental error to analyze performance of a PMS based on design intentions or desires. In this sense, we conclude that every PMS is producing only what it is capable of

producing, nothing more and nothing less. If the results are not consistent with expectations or desires, then we must first conclude that we have an incomplete representation and understanding of the technical and human entities, their relationships, boundaries, and transformation processes that are producing the output/outcome results.

PMS purpose may be specified in advance, however; only after operation of the system can achievement of the 'designed' purpose be confirmed. Beer [32] also adds that the system purpose is dependent on the perspective of the observer (stakeholder) of the system. Therefore, this implies that system purpose is observer dependent and should not be considered an *a priori* property, either in design or execution, for a PMS.

3.1.2. Complex system – a system is considered complex when it exhibits the following characteristics [9]:

- Large number of variables or elements
- Rich interactions among elements
- Difficulty in identification of attributes and emergent properties
- Loosely organized (structured) interaction among elements
- Probabilistic, as opposed to deterministic, behavior in the system
- System evolution and emergence over time
- Purposeful pursuit of multiple goals by system entities or subsystems (pluralistic)
- Possibility of behavioral influence or intervention in the system
- Largely open to the transport of energy, information, or resources from/to across the system boundary to the environment

These characteristics are fundamental to the thinking about design, deployment, operation, and transformation of PMSs. The importance of this concept for PMS is that treatment as a complex system requires acceptance of the limitations in our ability to understand, articulate, represent, and influence complex systems to achieve desired results. Instead, PMS design must be continuous and respect the probabilistic, dynamic, and emergent nature of complex systems.

3.1.3. Self-organization - holds that most of the structural and behavioral properties of a system emerge through interaction of the system elements [34]. Therefore, the actual design of a system can only be partially specified in advance of system operation. From the systems perspective, this explains why the most thoughtful and carefully designed PMSs have unintended consequences once deployed. In essence, system behavior and informal structure emerge only

through system operation, regardless of the detailed design efforts conducted prior to system deployment. Effective design of complex systems ensures that only the minimal essential constraints, necessary to achieve the desired level of output/outcome performance, are imposed on the system. In systems theory this concept of minimal constraint is also known as minimum critical specification [34] and [35]. Overspecification of system requirements: (1) is wasteful of scarce resources necessary to monitor and control system performance, (2) reduces system autonomy which in turn restricts the agility and responsiveness of the system to compensate for environmental shifts, and (3) fails to permit subsystem elements to self-organize based on their contextual knowledge, understanding, and proximity to the operating environment. Therefore, self-organization suggests that PMSs should specify only the minimal requirements necessary to achieve desired project results.

3.1.4. Complementarity – suggests that any two different perspectives (or models) of a system will provide different knowledge about the system. This knowledge is not entirely independent nor is it entirely compatible [33]. Each system perspective is correct from a particular vantage point of the system. In addition, each system perspective may also be considered, to some degree, to be incorrect from an alternate system vantage point. The important argument is that there are multiple system vantage points, each adding to a more holistic impression of the system. Shifts in vantage points, environmental conditions, or knowledge will influence perspectives of a system as well as its performance. It is naive to consider there is only one system perspective that is “correct”. Therefore, it is a mistake to conduct inquiry as to which system perspective is ‘right’. Assumption of a ‘right’ system encourages advocacy and competition instead of dialogue and collaboration.

3.1.5. Dynamic stability – a system remains dynamically stable as long as it can continue to produce required performance during environmental turbulence and changing conditions. Maintenance of stability, or dynamic equilibrium [36], in complex systems is achieved through adjustments to shifts and disturbances (internally or externally generated) that impact system performance. Neither PMSs nor their environments remain static and free from change. Therefore, maintenance of stability must be a continuous struggle to adjust and compensate for shifts. Stability of PMSs must be achieved through robust designs that provide for dynamic stability over a wide range of fluctuating conditions. This occurs through system environment scanning, internal monitoring, and feedback to provide for continued dynamic stability over the life cycle of the project.

4. A systems based model for PMSs

Exhibit 1 is our representation of the structure for a *PMS*. The systems perspective concludes, "Every system and organization has an architecture, or 'structure' broadly defined which largely determines what the system can and can't do [31, v] ". Therefore, system structure is a critical concern when considering improving effectiveness of a system.

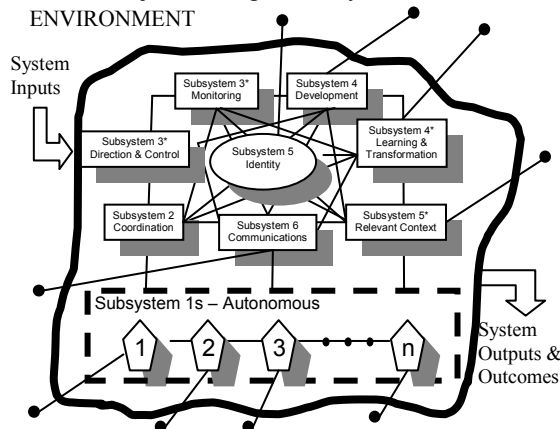
In the systems view, structure is "the set of arrangements by which the resources of an organization, human and others, are connected through relationships [32, p.20]." These relationships occur through mechanisms of interaction. For example, a weekly project staff meeting would be considered a structural mechanism. The meeting allows relationships between individuals and the entities they represent to occur. Thus, the systems view suggests that structure includes the system entities (technical, human), their specific implementing mechanisms (formal and informal), and the relationships that emerge through interaction in context. The arrangement of the total set of formal and informal mechanisms, coupled with the emergent relationships, constitutes the ‘system structure’. Therefore, understanding mechanisms that facilitate relationships is key to understanding system structure.

The Viable System Model (VSM) [31, 32, 33] offers a comprehensive systems perspective of structure. This model was developed from the field of cybernetics and systems theory [34]. The thrust of the model is a set of functions that exist in every operational system and are essential to maintain viability. A system maintains viability if “it is capable of responding to environmental changes even if those changes could not have been foreseen at the time the system was designed” [9, p.105]. The two primary VSM concerns with respect to structure are: (1) the arrangements of the structural entities that perform the essential functions of a viable system, and (2) the information flow (communications) between the different structural entities. In consideration of these two primary concerns the VSM has proven to be an effective model. It has been successfully applied to understand system structure in a variety of organizational settings ranging from industry to non-profit enterprises [37]. The VSM provides a foundation that can be used to establish the architecture for the PMS.

The foundation of the VSM is the identification of five primary management systems that must perform the functions essential for any viable system to maintain existence. Beer [32] suggests that viability is the ability to maintain a separate existence, which is achieved through effective system structure. In taking a systems view, system structure is the key to viability

and is taken as both the physical entities as well as their relationships. The five systems of the VSM [31, 32, 33] for viability include: System 1 (operations), System 2 (co-ordination), System 3 (control), System 3* (Audit), System 4 (intelligence), and System 5 (policy). Each of the five systems performs a unique and vital function in providing for the continued existence of the system. The degree to which the operation effectively performs these functions determines the efficacy of the structure and ultimately the level of system performance. An understanding of these systems provides insight for defining system structure.

Exhibit 1. Project Management System Model



Based on the seminal work of Beer, [31], [32] and [33], and supplementary expositions by [9], [37], [38] and [39], the Viable System Model has been used as a foundation for development of our PMS model.

The PMS model is based on nine subsystems responsible for transformation of input to output/outcomes. The nine PMS “subsystems” include:

Subsystem 1 - Autonomous Entities - This subsystem functions to produce the deliverables of the project in support of project objectives. In effect, these are products and services that meet a customer need or solve a problem. The other subsystems of the PMS exist to support and enhance the capabilities of Subsystem 1s to effectively produce project deliverables. There can be multiple Subsystem 1s in the PMS. Subsystem 1s can be added or eliminated throughout the life cycle of a project. Autonomy is a function of the relative independence of decision and action in a system. An example of a Subsystem 1 would be a major work breakdown structure element.

Subsystem 2 – Co-Ordination - This subsystem provides system stability by synchronizing the different subsystems. Unnecessary system oscillation is dampened among operational entities (System 1s) as well as the other Subsystems. System 2 prevents

unnecessary oscillation due the potentially uncoordinated activities of the other subsystems. For example, standard procedures, schedule planning groups, and standards committees are examples of technical mechanisms designed to accomplish System 2 functions.

Subsystem 3 – Direction & Control - Subsystem 3 functions to maintain operational performance on a day-to-day basis. As such, this subsystem provides for the execution of policy, allocation of resources, and accountability. The focus of Subsystem 3 is on internal operation of the project. Traditional day-to-day management of project operations, including activities such as near term planning, operational decisions, and resource allocation would be considered Subsystem 3 activities. For example, making decisions concerning project schedule modifications would be a Subsystem 3 activity.

Subsystem 3* - Monitoring – Related to Subsystem 3 is a separate subsystem to provide monitoring of operations for problems or deviations in performance. The primary function of System 3* is performance monitoring, which includes auditing of operations for performance assessment or analysis of deviations (*read System 3 Star). System 3* performs both routine and sporadic audits of problematic areas. For example, review of project status or a safety investigation are both examples of performing a Subsystem 3* monitoring activity.

Subsystem 4 - Development – System 4 captures environmental information and processes that information for operational impact and strategic implications. The focus for this system is directed externally and to the future, primarily concerned with gathering and processing environmental intelligence. Strategic project planning, environmental scanning and new technology evaluations are examples of Subsystem 4 activities.

Subsystem 4* - Learning & Transformation – Related to Subsystem 4, this subsystem provides for purposeful learning in the project system and is essential to effective transformation of the system based on shifts in the operational context and environment. Subsystem 4* learning is concerned with errors that are rooted in an inadequate PMS design, as opposed to faulty execution of the design. Errors are produced when the expected results of system operation are incongruent with what actually happens. While Subsystem 3* is focused internally on deviations in operational performance, Subsystem 4* questions errors which may be a consequence of a faulty system design. The correction of errors identified by Subsystem 4* requires modification in the PMS. For

example, a project scheduling process that is being followed precisely but yet is still producing less than desirable results would be a candidate for a Subsystem 4* inquiry.

Subsystem 5 - Identity - This subsystem assumes responsibility for the strategic decisions and direction that maintain the identity of the system. Identify is the set of unique characteristics and qualities that are the essence of the project, within which the subsystems of the PMS must operate. Elimination of one or more of these identity characteristics would result in the project losing distinctiveness and clarity. Additionally, this subsystem monitors and maintains a balance between the long-term external focus of Subsystem 4 and the short-term internal focus of Subsystem 3. The opposite foci of Subsystems 3 and 4 create a natural and predicted tension. Subsystem 5 must monitor and direct appropriate balance and resource allocation between present operations and future development. Establishment of long-term vision, mission clarification, strategic decisions, and project policies are examples of Subsystem 5 activities.

Subsystem 5 - Relevant Context* - This subsystem provides for identification and processing of the context. Project context is the relevant set of circumstances, factors, conditions, and patterns that are influential in constraining and enabling the PMS. The term *relevant* captures the necessity to reduce, from the infinite number of contextual possibilities, to a set that will be actively managed. Every PMS must operate within a context. Exhibit 2 identifies several key areas for consideration in identifying the relevant context for a PMS. Although the listing is not complete, it does provide a starting point for examination and articulation of relevant PMS context. Every project structure is embedded in a context that both constrains and supports the project. A project system structure cannot be understood independent of its environment or operational context.

Subsystem 6 - Communications - This subsystem provides for the flow of information through the PMS. The Communication Subsystem is responsible for establishment, monitoring, and transformation of the mechanisms that provide for the right information, to the right place, at the right time. In addition, design and execution of the mechanisms for dialogue and interpretation are important functions of this subsystem.

5. System environment - System environment includes the conditions, stakeholders, entities, and patterns external to the system that either have an influence on the system or are influenced by the system. The relationship of structure to the

environment is one of mutual response and fit. If the environment shifts in ways that affect system performance, then the structure must compensate to maintain stability of the system. This permits the system to maintain performance in the midst of change. In addition, notice that not every subsystem is linked to the environment. Subsystems 2 and 3 are not environmentally linked. This is because their function is entirely focused on the internal operations of the PMS.

6. Case study application The concept and model, of a systems science PMS perspective, was applied to a complex system of systems (SoS) project. In general, the SoS project encompassed multiple government agencies, regulatory bounded to deliver an overarching meta-level security system. In an effort to achieve the meta security system, each agency applied their unique and distinct autopoietic, self-maintaining, unity system towards a new functioning allopoeitic - development of a new process other than the individual ones - Meta system.

Exhibit 2. Areas for Relevant Context Development.

CONTEXTUAL ELEMENT	DESCRIPTION
Vision, Mission/Objectives	The idealized direction of where the higher level system (organization) wants to transition. Established PMS external direction and constraint
Strategy	The approach(es) for establishing direction to accomplish system objectives and provide direction for operational action.
Priorities	The primary organizational concerns to which scarce operational resources are directed and the project of interest must support.
Performance	The external metrics used to determine adequacy of project system performance.
Support Systems/Processes	Activities and external functions of the organization that both constrain and enable PMS autonomy and performance.

Management Style	The distinctive ways in which management presence influences project operations.
Staff Skills	The availability of recognized competencies necessary to perform the primary work of the project.
Reward/incentive	The recognition, formal and informal, for performance of individuals, units, and the overall project. These are both internal and external to the project.
Decision Making	The formal and informal methods for making external decisions that impact the project system.
Work Climate & Culture	The state of working conditions within the project system. Internal and external patterns that define acceptable approaches for decision, actions, and expectations for individual and collective behavior in the system. These patterns both constrain and enable performance in the project system.

This new meta system was formed within a unique environment of regulatory requirements, a lack of formal required interagency hierarchical reporting structures, self-organizing volunteer participation, and driven by personal relationships and high levels of personal desires for success. Exhibit 3 overviews the relevant context development of project. The applied systems based PMS followed the discussed PMS model as depicted in Exhibit 1 and outlined in Exhibit 4. The project output consisted of a final report that was to (1) identify and describe the system within its context and environment, (2) provide an analyses of the current system capabilities, and (3) provide a forward *metasystem* look towards system enhancements with potential performance improvements.

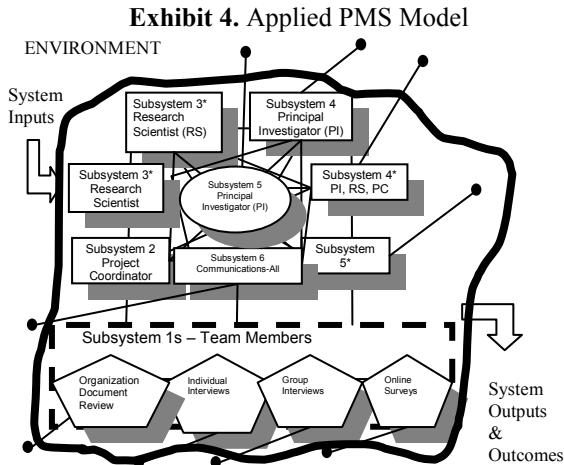
To achieve this effort the project team incorporated the technological approaches such as the Viable Systems Model, meetings, and project schedules, to bring

structure and order for project performance. The human component of the project involved the core project team and numerous government agencies' stakeholders that operated, at times as individuals, and at other times as collective group.

Exhibit 3. Areas for Relevant SoS Context Development

CONTEXTUAL ELEMENT	DESCRIPTION
Vision, Mission/Objectives	Establishment of a meta security system that provides close interagency coordination and rapid response
Strategy	Meta system approach is the utilization of individual autopoietic operating systems within a broader allopoeitic context to develop the meta system
Priorities	Competing organizational priorities of personnel and funding resources to accomplish the individual organizational mission and the new meta system
Performance	Establishment of new metrics to determine the meta system adequacy of performance
Support Systems/Processes	Competing and supporting internal and external functions, requirements, and stakeholder views that both constrain and enable the system
Management Style	Self-organizing, relationship based, and non-hierarchical management structure fostering cooperation without any overarching authority
Staff Skills	Overall each organization staffed with highly skilled, knowledgeable, and capable individuals
Reward/incentive	Internally driven reward systems based on the desire to do the best
Decision making	Decision making founded on cooperation and consensus agreements
Work Climate &	A context based on

Culture	cooperation and volunteerism with a predominate level of ascribed, versus delegated or expertise power
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Analysis of the project results and performance can be judged from two perspectives. First, did the project deliver the agreed on project output within the agreed on time frame and allocated budget? These are viewed as the traditional project performance measurement metrics. Based on this set of metrics the project was successful in that the final report was provided within the allocated time line and within the allocated budget. Based on these metrics the system engineering based PMS was successfully applied.

A second measure of project performance looks within the project, itself, an analyses the processes involved in delivering the final output. This analysis is commonly referred to as 'lessons learned.' The purpose of this assessment measurement is to identify those methods, tools, techniques, and process that worked well, and should be duplicated, and those that didn't work well and should be modified in future efforts. The case study PMS, as anticipated, encountered both positive and negative implementation challenges due to the system principals of (1) system purpose, (2) complex system, (3) self-organization, (4) complementarity, and (5) dynamic stability. System purpose was a major challenge due to the contextual setting that relied upon personal interactions, lack of a hierarchical required authority level, and the need to balance individual organizational operational requirements and the new *meta* system needs. The inherent conflicts were demonstrated by the varying stakeholder views of the *meta* system purpose. The *meta* system is a complex system as it is rich with interactions, loosely organized, and probabilistic in

nature. The system is also very open to the environment with a lack of clear boundaries. Complexity hampers the PMS application as emerging properties are routinely encountered and must be handled on a case by case basis. The system is also self-organizing with the individual organizational desires, needs, and wants, coupled with the associated personalities were the driving forces behind the system. As the individual alliances were built, modified, and changed the system was impacted both positively and negatively.

Complementarity was evident throughout the process as each stakeholder and various project team members had their system view. Often the views overlapped yet each view was unique based on the perspective of the individual.

Dynamic Stability has occurred within the system as the meta system is in place and operating. System changes occurred in response to emergent events, system changes and organizational focus. As the need to change arose the system adapted. Overall, Exhibit 5, identifies some prime lessons learned. The case study identified that in this unique situation a system based PMS is a viable and applicable method. Within the context of a complex system of systems, a PMS did improve the probability of project success. The system view allowed the project team to dynamically apply patterns of decision, action, and interpretation that enhanced the opportunity for project success. The case study also supports the view that further research and field based application analysis is required.

The project output consisted of a final report that was to (1) identify and describe the system within its context and environment, (2) provide an analysis of the current system capabilities, and (3) provide a forward metasytem look towards system enhancements with potential performance improvements.

Exhibit 5. Prime Lessons Learned

Maintain a flexible schedule	While the end date did not change milestone dates did change due to resource limitations and evolving understanding of the system.
Keep the PMS adaptable	While the project started with one PMS as the project evolved changes were required.
Ensure a better understanding of the PMS system	Technology, such as the VSM, can hinder or assist. If the stakeholders or project team members do not fully grasp the technology it hinders the discussions and impedes project progress.

Focus more attention on developing the stakeholder common terminology understanding	A common terminology assists while a lack of a common terminology hinders. More effort was required to build the common terminology between the stakeholders and project team.
Apply greater efforts in developing a converging stakeholder worldview	Each stakeholder views the new meta system through their unique worldview. This created the situation of many worldviews that hindered forward progress to a common meta system worldview
Acceptance of emergence and change	Emergence and change were a common occurrence. Plans were altered due to unforeseen circumstances, changing context and environmental impacts. Emergence and change are the norm not the exception.

7. Conclusions and implications

In this paper we have developed the concept of PMS from the perspective of systems science. PMS was defined as:

The structured set of technical and human entities that interact both formally and informally within a specific context. The products of interaction are patterns of decision, action, and interpretation that ultimately produce project results.

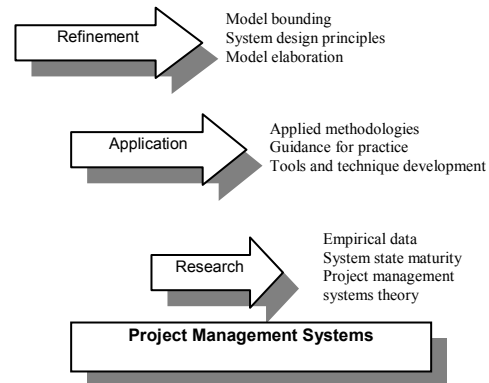
In addition, a systems based PMS model was presented. The model is important because it begins a more rigorous development of the concept of PMS. Although project management is maturing as a discipline, it still lacks in the application of systems science to better understand phenomena associated with design, deployment, operation, and transformation of PMSs.

With the PMS model developed, there are several implications for extending the current work. There are three primary areas that need to be addressed as the PMS model moves forward. These areas are identified as model refinement, application in practice, and research.

This is our first articulation of the PMSs model. Although we are confident in the fundamental structure and basis for the model, it requires further refinement and elaboration. In addition, the bounding of the model with respect to applicability must be developed.

Is the model appropriate to all classes of projects or just limited classes? Does the model apply equally for projects in different phases of their life cycle as well as initial design? Are there inherent limitations in the model based on the systems perspective? These are several areas that must be considered in refinement of the PMS model (Exhibit 6).

Exhibit 6. Further Development of PMSs.



A second expansion area is application development. The model cannot be deployed without an accompanying methodology to support structured application. Lacking a supporting methodology, the model is likely to be deployed in ways that do not offer the greatest utility and probability of success. Also, providing specific guidance, tools, and techniques for practice will assist practitioners in finding utility in the model. The focus on specific guidance must stay clear of unnecessary foundations development while providing enough depth for application. For practice, the degree of “systems grounding” necessary to effectively deploy and derive gain from the “application side” of the model needs to be explored. We do not expect to require practitioners to become PMSs scientists to gain benefits from applications derived from the model. However, the question for development is: *To what degree do users of the model require grounding in systems theory to gain practical utility?*

Finally, the model provides a foundation for systems research in project management. There are three primary avenues that offer research promise for PMSs. First, *design and demonstration of PMSs methodologies*. Research based methodologies must be developed that provide for design, deployment, operation, and transformation of PMSs. This research direction must establish empirical data concerning the deployment of systems based methodologies. Second, *research into the “maturity” of PMSs* is an essential direction for future efforts. There has been significant discussion concerning “project management maturity” [12] and [40]. In fact,

many models exist for maturity as pointed out by Andersen and Jessen [41]. Project Management Institute (PMI) has kept supporting the improvement of Organizational Project Management Maturity Model (OPM3[®]). However, the current models attempt, i.e. Capability Maturity Model by Software Engineering Institute, to benchmark the software process in order to establish a criterion to assess the organizational project management maturity and determine key areas to continuously improve project management practices. Research directed at *PMS state maturity beyond the software processes* is an important direction for PMSs. Finally, as a discipline, Project Management is *lacking in theoretical development*. Although this may be due to the “pragmatic and transdisciplinary nature” of the project management discipline, there is a need for inductive research to begin formation of PMSs theory.

8. References

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